**Academic Year: 2026-27** **Semester: V**
**Class/Branch: TE/DS** **Subject: Agile**

## LEVEL UP ASSIGNMENT

**Course Outcome(CO)/Lab Outcome(LO):**

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| LO1 | To apply Agile and Scrum methodologies using Jira for sprint planning, backlog management, and progress tracking, and integrate Agile workflows with DevOps pipelines. |
| LO2 | To apply Git for distributed version control, source code management, branching and merging strategies, and collaborative team development. |
| LO3 | To build and manage automated CI/CD pipelines using Jenkins and GitHub Actions for building, testing, packaging, and deploying applications. |
| LO4 | To Containerise, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and provision cloud infrastructure using Terraform. |
| LO5 | To implement ML lifecycle management and observability using MLflow, Airflow, AWS Lambda, Prometheus, and Grafana. |
| LO6 | To Design, integrate, and demonstrate an end-to-end DevOps pipeline covering planning, build, test, deployment, and monitoring. |

**Q 1 :** Implement the Scrum framework for a selected software development project (e.g., Library Management System / College Event Management System) using Jira by performing the following activities:

1. Create a Scrum project in Jira and define the Scrum team roles (Product Owner, Scrum Master, and Development Team).
2. Identify project requirements and create User Stories with suitable Acceptance Criteria.
3. Create and prioritize the Product Backlog based on business requirements.
4. Conduct Sprint Planning by selecting User Stories and preparing a Sprint Backlog with task assignments and timelines.
5. Simulate Scrum activities by updating task status, conducting Daily Scrum discussions, and tracking sprint progress using a Scrum Board.
6. Demonstrate Sprint Review and Retrospective activities and prepare suitable Scrum artifacts, including the Product Backlog, Sprint Backlog, and Increment.

**Q 2** : Select a software development scenario such as an Online Food Ordering System, Student Attendance System, or E-Commerce Application. Choose any three Agile development models from the following:

- Extreme Programming (XP)
- Feature Driven Development (FDD)
- Crystal
- Kanban
- Adaptive Software Development (ASD)
- Test Driven Development (TDD)

- Lean Software Development

For each selected model, perform the following activities:

1. Identify the key practices and principles applicable to the selected project.
2. Prepare a workflow or process diagram representing the implementation of the model.
3. Demonstrate the selected model's practices through suitable project activities.
4. Implement at least one practical activity, such as pair programming, test-first development, feature-based planning, Kanban board configuration, or waste identification.
5. Document the outcome with screenshots, sample outputs, or a demonstration of the implemented workflow. [L3]

**Q 3**: Consider a software development project in which customer requirements change frequently, development resources are limited, and stakeholders provide continuous feedback. Select a suitable project scenario and perform the following activities:

1. Prepare an initial set of project requirements and create a Product Backlog.
2. Simulate a change in customer requirements during development and update the Product Backlog accordingly.
3. Prioritize requirements based on business value, urgency, and available resources.
4. Plan a short development iteration by assigning tasks to available team members.
5. Demonstrate stakeholder collaboration through a simulated customer feedback or review session.
6. Implement changes based on feedback and demonstrate the updated working Increment.
7. Prepare a brief report explaining how the applied Agile practices helped manage changing requirements, resource constraints, and customer expectations.[L3]